

Conformal predictive intervals in survival analysis: a resampling approach

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ABSTRACT

The distribution-free method of conformal prediction has gained considerable attention in computer science, machine learning, and statistics. Candès et al. extended this method to right-censored survival data, addressing right-censoring complexity by creating a covariate shift setting, extracting a subcohort of subjects with censoring times exceeding a fixed threshold. Their approach only estimates the lower prediction bound for type I censoring, where all subjects have available censoring times regardless of their failure status. In medical applications, we often encounter more general right-censored data, observing only the minimum of failure time and censoring time. Subjects with observed failure times have unavailable censoring times. To address this, we propose a bootstrap method to construct 1- as well as 2-sided conformal predictive intervals for general right-censored survival data under different working regression models. Through simulations, our method demonstrates excellent average coverage for the lower bound and good coverage for the 2-sided predictive interval, regardless of working model is correctly specified or not, particularly under moderate censoring. We further extend the proposed method to several directions in medical applications. We apply this method to predict breast cancer patients' future survival times based on tumor characteristics and treatment.

KEYWORDS: bootstrap sampling; conformal prediction; predictive interval; right censored data.

1 INTRODUCTION

Conformal prediction has garnered significant attention and continues to grow in importance for estimating prediction intervals without relying on correctly specified model assumptions between outcomes and covariates (Vovk et al., 2005; Angelopoulos and Bates, 2021). In cancer research, there is a strong interest in providing predictive intervals for survival outcomes based on patients' baseline risk factors and treatments received. Inflammatory breast cancer (IBC) is a rare subtype of all breast cancer cases, characterized by aggressive cancer cells that block lymph vessels in the breast skin. IBC often progresses rapidly, with all cases being diagnosed as stage III or IV disease, carrying a poor prognosis. Age, race, tumor characteristics, and treatments received after diagnosis are important risk factors associated with the survival of IBC patients. Predicting the prognosis or survival time of an IBC patient is crucial for both the patient and her physicians in selecting the most appropriate initial course of treatment and managing the disease optimally. While it is widely recognized that tumor characteristics, such as nodal status and stage at diagnosis, play a significant prognostic role in predicting long-term outcomes for non-IBC breast can-

cer patients, the impact of these tumor characteristics and demographic factors on the prognosis and response to established therapies in IBC patients has not been well-studied. Existing literature suggests that all subtypes of IBC have poor overall survival and a worse prognosis compared to their non-IBC counterparts of the same subtypes (Masuda et al., 2014). Accurately predicting overall survival time based on tumor characteristics and treatment information using the largest IBC dataset, the National Cancer Database (NCDB), could generate hypotheses for future prospective trials aiming to improve the overall survival of IBC patients.

Building on recent conformal inference literature, we propose a conformal predictive interval (CPI) for right-censored data in regression settings. One major advantage of the method is that it does not require a correct model assumption for the conditional distribution of $(T|\mathbf{X})$ where T represents the survival outcome and $\mathbf{X}$ is a set of covariates. In contrast, conventional predictive models rely on a correct model assumption of the outcome given the covariates $(T|\mathbf{X})$. The essence of "model assumption free" in conformal inference is that the conformal prediction interval always has the correct average coverage probability under a work-

ing regression model. The average coverage probability refers to the probability that the true value of the survival time falls within the predictive interval.

2 PRELIMINARIES

2.1 Traditional predictive intervals

The prediction interval projects a range in which the outcome of a future patient will fall given a set of covariates, while a conventional confidence interval estimates the possible range of the statistical parameter of the data, such as the population mean. Suppose we have a training sample with independent and identically distributed (iid) data $(Y_i, \mathbf{X}_i)$, $i = 1, 2, \dots, n$, where Y_i is a response variable and $\mathbf{X}_i$ is a p -dimensional vector. It is of interest to predict a future response Y_{n+1} given a new set of covariates $\mathbf{X}_{n+1}$.

In the traditional prediction confidence approach, one may first postulate a parametric conditional density model $f(y|\mathbf{X} = \mathbf{x}; \boldsymbol{\theta})$ in a regression model. The unknown p -dimension parameter $\boldsymbol{\theta}$ associated with corresponding $\mathbf{X}$ can be estimated by using the training samples, denoted as $\hat{\boldsymbol{\theta}}$. A predictive set $\hat{C}(\mathbf{x})$ can be obtained such that

$$P(Y_{n+1} \in \hat{C}(\mathbf{x}) | \mathbf{X}_{n+1} = \mathbf{x}) \geq 1 - \alpha \quad (1)$$

approximately for a given mis-coverage probability of α (Geisser, 1993). The methods based on the plug-in estimate $\hat{\boldsymbol{\theta}}$ have been criticized for ignoring the uncertainty of $\hat{\boldsymbol{\theta}}$ in the estimated confidence interval $\hat{C}(\mathbf{X})$. Specifically, both the variation of the regression parameter estimate and the variation of the error term of the future observation should be considered (Lawless and Freddette, 2005).

2.2 Conformal predictive interval

Vovk et al. (2009) introduced a seminal concept of conformal prediction, which recently has been utilized in statistical inference with novel machine learning algorithms (Lei et al., 2018). The conformal prediction considers data uncertainty from a working model, and constructs a statistically valid prediction interval for a future outcome Y_{n+1} given covariate $\mathbf{X}_{n+1}$ and training data. Using the same data notation as above, the CPI is defined as

$$P(Y_{n+1} \in \hat{C}(\mathbf{X}_{n+1})) \geq 1 - \alpha, \quad (2)$$

where the probability is with respect to the joint distribution of $(Y_{n+1}, \mathbf{X}_{n+1})$ rather than to the conditional distribution when constructing a traditional predictive interval in (1).

One primary advantage of CPI is that the conditional density of Y given $\mathbf{X} = \mathbf{x}$, $f(y|\mathbf{x}; \boldsymbol{\theta})$ serves as a “working model” for the relationship between the outcome and the covariates. Regardless of whether the working model is correct, the overall coverage probability of the conformal intervals remains at the specified level asymptotically.

The CPI takes into consideration the variation of $\mathbf{X}$ and Y . Thus, the interval is generally wider than the conventional predictive confidence interval, which only involves the variation of Y for a fixed value of $\mathbf{X}$. For complex data structures (eg, with a high dimension of $\mathbf{X}$), conformal inference offers an effective ap-

proach to predicting outcome Y by leveraging cutting-edge machine learning techniques.

2.3 Conformal inference for survival data

Despite the success of developing conformal intervals for uncensored data, there are limited works in conformal inference for general right-censored data. Consider iid observed data as $Y_i = \min(T_i, C_i)$, $\delta_i = I(T_i \leq C_i)$, $\mathbf{X}_i$, $i = 1, 2, \dots, n$, where T_i , C_i are, respectively, the failure time and censoring time for individual i , δ_i is the censoring indicator, and $\mathbf{X}_i$ is a $p \times 1$ vector of covariates. Candès et al. (2023) proposed a covariates-dependent lower conformal prediction bound for survival data subject to Type-I censoring. Under the conditional independent censoring assumption, the following inequality holds:

$$\begin{aligned} 1 - \alpha &= P(T \geq q_\alpha(\mathbf{x})) \geq P(T \geq q_\alpha(\mathbf{x}))P(C \geq q_\alpha(\mathbf{x})) \\ &= P(Y \geq q_\alpha(\mathbf{x})), \end{aligned}$$

thus $P(Y \geq q_\alpha(\mathbf{x})) \leq 1 - \alpha$, where $q_\alpha(\mathbf{x})$ is a marginally calibrated lower probability bound of uncensored T given $\mathbf{x}$ at significance level $1 - \alpha$. If the CPI of T is constructed based on the observed $Y = \min(T, C)$, it would be too conservative, especially when censoring occurred early. Instead, Candès et al. (2023) introduced an artificial truncation at a specified value c_0 by considering only training data samples with their censoring times $C \geq c_0$. Under the assumption that T and C are independent given covariate $\mathbf{x}$, $P_{(\mathbf{X}, \min(Y, c_0)) | C \geq c_0} = P_{\mathbf{X} | C \geq c_0} P_{\min(T, c_0) | \mathbf{X}}$. On the other hand, $P_{(\mathbf{X}, \min(T, c_0))} = P_{\mathbf{X}} P_{\min(T, c_0) | \mathbf{X}}$. Thus, the distribution of $(\mathbf{X}, \min(T, c_0))$ in the subpopulation after right truncation and that of the whole population satisfies the so called covariate distributional shift (Tibshirani et al., 2019). While it is novel to utilize artificial truncation to the training data and to apply the method of conformal prediction under covariate shift (Lei and Candès, 2021), this approach is only applicable when the censoring time C is fully observed (Type-I censoring) for all individuals to construct the subpopulation from the training data.

In medical research, we encounter more general censored survival data, where the censoring times are not available for individuals whose failure event occurred before the random censoring event. Moreover, after excluding data observations with $C < c_0$ in Candès et al. (2023), the constructed CPI is based on the truncated outcome $\min(T, c_0)$ rather than the true survival time T . Thus, the interpretation of the estimator is different. Also, the choice of c_0 is tricky. A large value of c_0 results in a smaller subpopulation for training data, suffering more statistical efficiency loss. A small value of c_0 produces a less informative conformal interval. More importantly, as we will show in the motivating data analysis, 2-sided predictive intervals provide more critical information for medical decision making in practice than a 1-sided lower bound.

3 BOOTSTRAP CPIS

We propose an approach to constructing 2-sided as well as 1-sided CPIS for survival data with general censoring distributions, using the non-parametric bootstrap methods by Efron (1979). Denote the joint cumulative distribution function (CDF) of the

failure time and covariate as $F(t, \mathbf{x})$, and the conditional survival function of the failure time and censoring time as $S(t|\mathbf{x}) = P(T > t|\mathbf{x})$, $G(t|\mathbf{x}) = P(C > t|\mathbf{x})$. The key to the conformal inference is that one must find a non-parametric estimation of the joint distribution $F(t, \mathbf{x})$. The probability of observing a subject that failed at time $T = t$ with covariate $\mathbf{X} = \mathbf{x}$ (ie, $\delta = 1$) is given by

$$P(T = t, \mathbf{X} = \mathbf{x} | \delta = 1) \sim \frac{G(t|\mathbf{x})dF(t, \mathbf{x})}{P(C > T)}. \quad (3)$$

We propose a method to handle the general censored survival data when the shift function depends on the outcome. We first consider $G(t|\mathbf{x}) = G(t)$, that is, the censoring variable is independent of the covariates, and will present the general case where censoring distribution depends on $\mathbf{X}$ later. The joint CDF of $(T, \mathbf{X})$ can be consistently estimated by its empirical distribution function with an inverse weighted censoring distribution using the training data,

$$\widehat{F}(t, \mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \frac{\delta_i I(Y_i \leq t, \mathbf{X}_i \leq \mathbf{x})}{\widehat{G}(Y_i)}, \quad (4)$$

where $\widehat{G}(\cdot)$ is estimated by the Kaplan–Meier (K–M) method for censoring times. Note that the marginal distribution of T (or in the absence of covariates), the estimate in Equation (4), is reduced to $\widehat{F}(t) = \frac{1}{n} \sum_{i=1}^n \frac{\delta_i I(Y_i \leq t)}{\widehat{G}(Y_i)}$, which is precisely the K–M estimate of failure time (Satten and Datta, 2001).

To construct a CPI, it is common to utilize a working regression model for the failure time with given covariates. This model could be parametric or semiparametric. For the purpose of illustration, we use the Cox model as a working model without loss of generality. Denote the conditional survival function as

$$S(t|\mathbf{x}; \boldsymbol{\theta}) = \exp\{-\Lambda(t|\mathbf{x})\}, \quad \Lambda(t|\mathbf{x}) = \Lambda_0(t) \exp(\mathbf{x}\boldsymbol{\beta}),$$

where $\Lambda_0(t)$ is the baseline cumulative hazard function and $\boldsymbol{\theta} = (\boldsymbol{\beta}, \Lambda)$. We use the observed training data $(Y_i, \delta_i, \mathbf{X}_i)$, $i = 1, 2, \dots, n$ to estimate the unknown parameters by the partial likelihood method and the Breslow estimate, denoted as $\widehat{\boldsymbol{\theta}} = (\widehat{\boldsymbol{\beta}}, \widehat{\Lambda})$. Let $(\mathbf{X}_{n+1}, T_{n+1})$ be the testing data, where $\mathbf{X}_{n+1}$ is given, but T_{n+1} is to be predicted. We describe the step-by-step procedure to construct a CPI for T_{n+1} :

- **Step 1.** Generate the bootstrap resampling data, $\mathcal{D}^* = (Y_i^*, \delta_i^*, \mathbf{X}_i^*, i = 1, 2, \dots, n)$, from the original training data with equal probability and replacement.
- **Step 2.** Obtain $\widehat{\boldsymbol{\theta}}^* = (\widehat{\boldsymbol{\beta}}^*, \widehat{\Lambda}^*)$ under the working model with the bootstrap dataset, $\mathcal{D}^*$.
- **Step 3.** Randomly sample 1 pair $(T^b, \mathbf{X}^b)$ from the estimated empirical joint CDF, $\widehat{F}(t, \mathbf{x})$ in (4), which is estimated using the original training data.
- **Step 4.** Calculate $U^b = \exp\{-\widehat{\Lambda}^*(T^b) \exp(\mathbf{X}^b \widehat{\boldsymbol{\beta}}^*)\}$.
- **Step 5.** After repeating Steps 3–4 for B times, denote L and R as the low $[B\alpha/2]$ th and upper $[B(1 - \alpha/2)]$ th quantile of $(U^b, b = 1, \dots, B)$, respectively.
- **Step 6.** Construct the conformal interval by solving the following inequality for T_{n+1} :

$$L \leq \exp\{-\widehat{\Lambda}(T_{n+1}) \exp(\mathbf{X}_{n+1} \widehat{\boldsymbol{\beta}})\} \leq R, \quad \text{or}$$

$$\widehat{\Lambda}^{-1}\{-(\log R) e^{-\mathbf{X}_{n+1} \widehat{\boldsymbol{\beta}}}\} \leq T_{n+1} \leq \widehat{\Lambda}^{-1}\{-(\log L) e^{-\mathbf{X}_{n+1} \widehat{\boldsymbol{\beta}}}\}.$$

The proposed procedure under a general working model is described in **Algorithm 1**, in which $\boldsymbol{\theta}$ defines all unknown parameters of the working model.

Algorithm 1 Conformal prediction with right-censored data

Input: Training data $(Y_i, \delta_i, \mathbf{X}_i)$, $i = 1, \dots, n$, miscoverage level α , number of bootstrap replicates B , working model $S(t | \mathbf{x}; \boldsymbol{\theta})$, using the original training data estimate $\boldsymbol{\theta}, \widehat{\boldsymbol{\theta}}$, and the empirical estimator of the joint CDF of $(T, \mathbf{X})$, $\widehat{F}(t, \mathbf{x})$. Points $\mathcal{X}_{new} = \{\mathbf{x}_{n+1}, \mathbf{x}_{n+2}, \dots\}$ at which to construct prediction intervals.

Output: Prediction intervals at each element of $\mathcal{X}_{new}$.

Step 1: Generate the bootstrap data $\mathcal{D}^* = (Y_i^*, \delta_i^*, \mathbf{X}_i^*, i = 1, 2, \dots, n)$ from the original data with equal probability and replacement.

Step 2: Obtain $\widehat{\boldsymbol{\theta}}^*$ under the working model using $\mathcal{D}^*$.

for $b = 1, 2, \dots, B$ **do**

Step 3: Sample one pair $(T^b, \mathbf{X}^b)$ from the joint distribution $\widehat{F}(t, \mathbf{x})$.

Step 4: Calculate $U^b = S(T^b | \mathbf{X}^b; \widehat{\boldsymbol{\theta}}^*)$.

end for

Define L and R as the $[B\alpha/2]$ -th and $[B(1 - \alpha/2)]$ -th quantile among U^b , $b = 1, \dots, B$.

Return $\mathcal{C}_{conf}(\mathbf{x}) = [S^{-1}(R | \mathbf{x}; \widehat{\boldsymbol{\theta}}), S^{-1}(L | \mathbf{x}; \widehat{\boldsymbol{\theta}})]$ for each $\mathbf{x} \in \mathcal{X}_{new}$.

Note that the CPI under a semiparametric working model (eg, Cox model) cannot be extrapolated beyond the largest observed failure time $\eta = \max\{Y_i, \delta_i = 1, i = 1, 2, \dots, n\}$. In particular, when censoring is heavy in the training data, estimation of the predictive interval, especially the upper bound can be inaccurate. Instead, we can provide a more reliable predictive interval for the minimum of failure time T_{n+1} and η . The estimated conformal interval of $\min(T_{n+1}, \eta)$ is $[\widehat{\Lambda}^{-1}\{-(\log R) e^{-\mathbf{X}_{n+1} \widehat{\boldsymbol{\beta}}}\}, \min(\eta, \widehat{\Lambda}^{-1}\{-(\log L) e^{-\mathbf{X}_{n+1} \widehat{\boldsymbol{\beta}}}\})]$.

Note that the censoring distribution, $G(t|\mathbf{x})$, possibly depends on the failure time via the covariates, playing a role of shifting that results in a biased sampling version of $F(t, \mathbf{x})$. When predicting a failure time T with $\mathbf{X}$ but the observed training samples have censored outcomes, the covariate shift conformal inference method developed by Tibshirani et al. (2019) is not directly applicable to the setting with an outcome-dependent shift function. This issue is often referred to as prior probability shift or label shift in machine learning (Dockès et al., 2021). Conformal inference in survival analysis involves tackling a more intricate shift problem, wherein the shift function is contingent on both failure time T and covariate $\mathbf{X}$. This bears a resemblance to non-ignorable missing data problems, where the missing probability function $w(t, \mathbf{x})$. In our context, the shift function $w(t, \mathbf{x})$ is a function of censoring distribution $G(t | \mathbf{x})$, which always depends on T (and possibly on $\mathbf{X}$ too).

The following theoretical properties of the proposed conformal prediction interval ensure the average coverage probability to be valid asymptotically. Without loss of generality,

suppose the first n_1 observations are observed failures ($\delta_1 = \dots = \delta_{n_1} = 1$), and let $n_1 = \sum_{i=1}^n \delta_i$, and $\hat{\Delta} = \sum_{i=1}^{n_1} \hat{G}^{-1}(Y_i)$. We first establish the asymptotic behavior of the estimated CDF of the failure time and covariate, that is,

$$\tilde{F}(t, \mathbf{x}) = \frac{\hat{F}(t, \mathbf{x})}{\hat{F}(\infty, \infty)} = \sum_{i=1}^{n_1} \frac{\hat{\Delta}}{\hat{G}(Y_i)} I(Y_i \leq t, \mathbf{X}_i \leq \mathbf{x}). \quad (5)$$

Theorem 1: Under regularity conditions (C.1)–(C.4) in the [Supplementary Materials](#), $\tilde{F}(t, \mathbf{x})$ is a consistent estimate of $F(t, \mathbf{x})$, and $\sqrt{n}\{\tilde{F}(t, \mathbf{x}) - F(t, \mathbf{x})\}$ converges in distribution to a normal distribution with mean 0 and variance $\sigma(t, \mathbf{x})$, where $\sigma(t, \mathbf{x})$ is defined in the [Supplementary Material](#).

Theorem 2: Under regularity conditions (C.1)–(C.7) in the [Supplementary Materials](#), the bootstrap CPI is valid, ensuring correct coverage of $100(1 - \alpha)\%$ asymptotically.

4 SEVERAL EXTENSIONS FOR SURVIVAL CONFORMAL INFERENCE

4.1 Covariate-dependent censoring distribution

In medical research, the large training datasets often include multi-site patient samples with site-specific or the patients' covariate-specific censoring distribution. The method developed above can be generalized to the case when $G(t|\mathbf{x})$ depends on covariate $\mathbf{x}$ in (4).

When the censoring distribution is study-site dependent, we can estimate $G_k = P(C \geq t | S = k)$, $k = 1, 2, \dots, K$ for each study site separately, using the K–M estimate as $\hat{G}_k(t)$, where K is the number of study sites. Without loss of generality, $F(t, \mathbf{x})$ remains the same across sites. The joint CDF of $(T, \mathbf{X})$ can be consistently estimated by the inverse weighted empirical function

$$\hat{F}(t, \mathbf{x}) = \frac{1}{K} \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{\delta_{ki} I(Y_{ki} \leq t, \mathbf{X}_{ki} \leq \mathbf{x})}{n_k \hat{G}_k(Y_{ki})}, \quad (6)$$

where the observed data are denoted as $(Y_{ki}, \delta_{ki}, \mathbf{X}_{ki})$, $i = 1, \dots, n_k$, and $k = 1, \dots, K$.

When the censoring distribution depends on a set of covariates rather than a categorical variable, one could postulate a parametric or semiparametric regression model for C given $\mathbf{X}$. For example, under the Cox model for $G(t|\mathbf{X}) = G(t|\mathbf{X}; \boldsymbol{\gamma})$, the unknown parameters $\boldsymbol{\gamma}$ can be estimated by maximizing the likelihood

$$\prod_{i=1}^n [G(Y_i|\mathbf{X}_i; \boldsymbol{\gamma})]^{\delta_i} [dG(Y_i|\mathbf{X}_i; \boldsymbol{\gamma})]^{1-\delta_i}.$$

The joint CDF of $(T, \mathbf{X})$ can be consistently estimated by its empirical distribution function with the inverse weighted censoring distribution $\hat{G}(t|\mathbf{X}; \hat{\boldsymbol{\gamma}})$,

$$\hat{F}(t, \mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \frac{\delta_i I(Y_i \leq t, \mathbf{X}_i \leq \mathbf{x})}{\hat{G}(Y_i|\mathbf{X}_i; \hat{\boldsymbol{\gamma}})}. \quad (7)$$

In the simulation study, we evaluate how sensitive the empirical CDF (4) is if ignoring the covariate effect on censoring times, when the censoring times depend on covariates.

Theorem 3: Under regularity conditions (C.4)–(C.9) in the [Supplementary Materials](#), the bootstrap CPI with covariate-dependent censoring is valid, ensuring correct coverage of $100(1 - \alpha)\%$ asymptotically.

4.2 Conformal interval for remaining survival time

Another important practical clinical question is how we can predict a patient's survival time, given that the patient survived for c_L (eg, 2 years) after her initial diagnosis considering her tumor characteristics and treatment received. We can construct a predictive conformal interval for her remaining life expectancy, $T_{n+1} - c_L$, given that the individual survived c_L years with covariate $\mathbf{X}_{n+1}$, based on

$$P(T_{n+1} > t | T_{n+1} > c_L, \mathbf{X}_{n+1}) = \frac{S(t|\mathbf{X}_{n+1})}{S(c_L|\mathbf{X}_{n+1})}, \quad t > c_L > 0.$$

Interestingly, the above equation shows a covariate shift for the distribution of a subpopulation of patients who survived c_L years from that of the original whole population in the training data. We can modify the methods in the above Section to incorporate the intermediate information as follows:

- **Step 1.** Generate the bootstrap resampling data, $\mathcal{D}^* = (Y_i^*, \delta_i^*, \mathbf{X}_i^*, i = 1, 2, \dots, n)$, from the original training data with equal probability and replacement.
- **Step 2.** Obtain $\hat{\boldsymbol{\theta}}^* = (\hat{\boldsymbol{\beta}}^*, \hat{\Lambda}^*)$ by fitting the working regression model using the bootstrap data, that is, $\mathcal{D}^*$.
- **Step 3.** Randomly sample pair $(T^b, \mathbf{X}^b)$ from the conditional CDF, $\hat{F}(t, \mathbf{x} | t > c_L)$. This can be achieved by sampling pairs from $\hat{F}(t, \mathbf{x})$ but only keeping those $(T^b, \mathbf{X}^b)$ with $T^b > c_L$.
- **Step 4.** Calculate $U^b = S(T^b|\mathbf{X}^b; \hat{\boldsymbol{\theta}}^*)/S(c_L|\mathbf{X}^b; \hat{\boldsymbol{\theta}}^*)$, $T^b \geq c_L$.
- **Step 5.** After repeating Steps 3–4 for B times, obtain L and R as the low $[B\alpha/2]$ th and upper $[B(1 - \alpha/2)]$ th quantile of $(U^b, b = 1, \dots, B)$, respectively.
- **Step 6.** Construct the conformal interval by solving the following inequality,

$$L \leq \frac{S(T_{n+1}|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}})}{S(c_L|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}})} \leq R \text{ or equivalently}$$

$$S(c_L|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}})L \leq S(T_{n+1}|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}}) \leq S(c_L|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}})R.$$
 The $100(1 - \alpha)\%$ CPI is

$$S^{-1}\{S(c_L|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}})R|\mathbf{X}_{n+1}\} \leq T_{n+1} \leq S^{-1}\{S(c_L|\mathbf{X}_{n+1}; \hat{\boldsymbol{\theta}})L|\mathbf{X}_{n+1}\}.$$

Theorem 4 ensures the asymptotic coverage accuracy for the proposed conformal interval for remaining survival time.

Theorem 4: Under regularity conditions (C.1), (C.3)–(C.7), and (C.10) in the [Supplementary Materials](#), the bootstrap CPI for

$S(t|\mathbf{x})/S(c_L|\mathbf{x})$ is valid, ensuring correct coverage of $100(1 - \alpha)\%$ asymptotically.

5 SIMULATION STUDIES

We conducted extensive simulation studies to evaluate the performance of the proposed methods using various working regression models under different scenarios. We first assume the covariate vector included 4 components $\mathbf{X} = (X_1, X_2, X_3, X_4)$, where X_1 and X_3 were binary variables following a Bernoulli distribution with probability 0.5, X_2 and X_4 are continuous variables following a Uniform distribution on $U(-5, 5)$, and a standard normal distribution, respectively. The failure time T was generated either from the Weibull or log-normal distribution. The censoring time C was first generated independently from a uniform distribution $U(0, \tau_c)$, and τ_c was chosen to get the desired censoring rates of 15% or 50%. For each scenario, there were 2000 repetitions with a training sample size of 2000 and a testing sample size of $n_{test} = 2000$.

Using the proposed conformal inference method, a working regression model was needed. Though the working model does not have to be correct, we want to evaluate how the selection of a working model might impact efficiency and accuracy under different underlying failure and censoring distributions. Simulation results of the proposed CPIs will be summarized under 3 working models denoted as follows:

- (1) CPI-log-normal: $S(t|\mathbf{x}; \theta) \sim$ log-normal working model.
- (2) CPI-Weibull: $S(t|\mathbf{x}; \theta) \sim$ Weibull working model.
- (3) CPI-Cox: $S(t|\mathbf{x}; \theta) \sim$ Cox working model.

For comparison, the conventional conditional approaches (4)–(6) were also considered. These conditional approaches rely on the correct specification of the model. If the model was correctly specified, the conditional approaches would perform well. If a wrong model was used, the method may not be valid. We also compare the proposed method with 2 distribution-free approaches, a random forest model (7) using an ensemble of decision trees and a marginal K–M estimate ignoring covariates information (8).

- (4) Conditional log-normal model.
- (5) Conditional Weibull model.
- (6) Conditional Cox model.
- (7) Random forest model.
- (8) Marginal K–M estimation without using covariates.

For each of the conditional approaches (4)–(6), we first estimated $\hat{\theta}$ using the training data under the respective model assumed, and then for a new observation in the testing dataset, we obtain $\hat{S}(t_{n+1}|\mathbf{x}_{n+1}; \hat{\theta})$. The K–M estimate served as a reference without using covariate information in the testing dataset. For the 1-sided interval, the lower bound was calculated as the α th quantile from the estimated model; for the 2-sided inter-

val, the prediction interval was estimated from the $\alpha/2$ th and $(1 - \alpha/2)$ th quantiles of the conformal scores.

We evaluated the coverage rate of the prediction interval as $\sum_{i=1}^{n_{test}} I\{T_i \geq \hat{L}(X_i)\}/n_{test}$ for the 1-sided bound and $\sum_{i=1}^{n_{test}} I\{\hat{L}(X_i) \leq T_i \leq \hat{U}(X_i)\}/n_{test}$ for the 2-sided interval. Besides coverage probability, we also calculated the mean and standard deviation of the lengths of the prediction intervals as $\sum_{i=1}^{n_{test}} \{\hat{U}(X_i) - \hat{L}(X_i)\}/n_{test}$ for the 2-sided interval. Wider prediction intervals indicated less efficient estimates with higher uncertainty. A smaller lower bound value represented higher uncertainty for the 1-sided interval.

Table 1 and Figure 1 summarize the results from the 2-sided predictive intervals under 15% and 50% censoring rates. The proposed approaches under different working models (CPI-log-normal, CPI-Weibull, and CPI-Cox) all performed well, maintaining the specified coverage probability while having a reasonable prediction interval length when the censoring rate was low. The correct specification of working models did not impact the coverage rate or the length of the prediction interval much, although smaller variations were observed in the interval lengths. It is interesting to note that the average length of the CPIs under the Cox working model are much narrower and even better than these of CPI-Weibull and CPI-log-normal under the correct model assumptions. This is likely because the maximum support of CPI-Cox is at the largest observed failure time; the estimated CPI and its length are thus robust to the extreme failure times due to long-tailed parametric distributions. As expected, the conditional estimators are sensitive to the model assumption; they could lead to over or under-coverage if the working model is different from the true underlying distribution. When failure times follow log-normal distribution, the non-parametric K–M, random forest, and conditional Cox models were unable to maintain the specified coverage rate, 90%, for the 2-sided problem.

The simulation results with heavy censoring, 50%, showed that the proposed CPI with parametric working models performed well, achieving 89% coverage probability given the desired 90% coverage probability, whereas the CPI-Cox approach suffered an under-coverage issue because the survival distribution was not properly estimated beyond the maximum observed failure time when the support of the censoring distribution is shorter than that of the survival distribution. Note that the non-parametric confidence intervals using the K–M $F_{KM}^{-1}(\alpha/2)$ and $F_{KM}^{-1}(1 - \alpha/2)$, random forest, or the conditional Cox model suffered even worse coverage probabilities than CPI-Cox. If a model is misspecified as the log-normal model when failure times were generated from the Weibull distribution, we observed a wide prediction interval length with an overestimated coverage rate. Conversely, when data are generated from the log-normal distribution, the conditional log-normal approach can achieve coverage probability as expected with a smaller variation in the prediction interval length. In the presented scenarios, CPI-log-normal and CPI-Weibull achieved shorter average interval lengths compared to the conditional CIs under even correctly specified distribution (eg, 136 or 137 compared with 151 under log-normal).

In summary, if censoring is light in the training dataset, then a CPI with the Cox working model is recommended for its esti-

TABLE 1 Empirical 2-sided 90% coverage of the failure times.

Censoring	Distribution	Method	Cov	Failure times		Cov	min(failure times, η)	
				Length	SD		Length	SD
15%	Weibull	CPI-log-normal	0.90	21.35	1.67	0.90	13.82	1.02
		CPI-Weibull	0.90	20.84	1.55	0.90	13.53	0.97
		CPI-Cox	0.90	11.98	0.51	0.90	10.41	0.52
		Log-normal	0.93	31.08	2.35	0.93	20.11	1.44
		Weibull	0.90	21.52	1.35	0.90	13.97	0.84
		Cox	0.86	12.07	0.50	0.90	10.55	0.54
		Random forest	0.90	14.97	0.81	0.95	13.57	0.81
		K-M	0.90	46.40	3.58	0.95	46.40	3.58
	log-normal	CPI-log-normal	0.89	152.82	21.00	0.90	50.56	6.20
		CPI-Weibull	0.89	147.48	20.93	0.90	49.12	6.05
		CPI-Cox	0.90	29.59	1.43	0.90	24.70	1.47
		Log-normal	0.90	169.58	19.53	0.90	56.10	5.38
		Weibull	0.93	152.98	18.66	0.94	50.95	5.10
		Cox	0.86	31.76	1.42	0.94	25.67	1.52
		Random forest	0.87	37.91	2.45	0.96	32.64	2.35
		K-M	0.87	119.21	0.84	0.95	119.21	0.84
50%	Weibull	CPI-log-normal	0.89	50.88	6.28	0.90	7.53	0.79
		CPI-Weibull	0.89	50.39	5.46	0.90	7.49	0.74
		CPI-Cox	0.85	3.88	0.20	0.90	3.60	0.15
		Log-normal	0.94	87.01	10.61	0.92	12.88	1.34
		Weibull	0.90	54.46	5.26	0.89	8.09	0.71
		Cox	0.58	3.55	0.11	0.89	3.62	0.13
		Random forest	0.59	4.13	0.23	0.93	4.26	0.20
		K-M	0.60	8.90	0.01	0.92	8.90	0.01
	log-normal	CPI-log-normal	0.89	137.29	24.62	0.90	3.06	0.44
		CPI-Weibull	0.89	136.36	34.71	0.90	3.05	0.46
		CPI-Cox	0.72	1.00	0.05	0.90	1.09	0.06
		Log-normal	0.90	171.29	26.89	0.90	3.81	0.43
		Weibull	0.92	151.00	36.14	0.94	3.38	0.40
		Cox	0.56	1.11	0.04	0.93	1.16	0.06
		Random forest	0.56	1.20	0.08	0.94	1.37	0.08
		K-M	0.55	2.99	0.00	0.92	2.99	0.00

mation accuracy (much narrower CI) and robustness compared with the CPIs under parametric working models. In contrast, if censoring is heavy, the upper bound of CPI can be inaccurate for CPI with a semiparametric working model or non-parametric methods. In this case, using a parametric working model over the semiparametric model for CPI is preferable to ensure a reasonable coverage probability. Alternatively, we can provide a more reliable predictive interval for the minimum of failure time T^* and η , which is the maximum of observed failure times in the training data, regardless of censoring rate, as shown in the second column of Figure 1. Note that the length of CPI was much narrower for this estimate compared with its counterpart with a parametric working model.

In the [supplementary materials](#), Table S1 and Figure S1 summarized the results from the performance of 1-sided predictive intervals with heavy censoring of 50%. Though all methods performed relatively well, our proposed methods under 3 working models all maintained the desired coverage probability at the specified level, 95%, and with similar lower bounds. Note that heavy censoring has much less impact on the 1-side CPI than the 2-sided CPI, since right-censoring mainly affected the estimation of the upper bound of a predictive interval. The conditional approach led to under or over-coverage under 2 underlying distributions. The K-M estimator had more severe under-

coverage and smaller low bounds than those from random forest method.

Using $B = 1000$ bootstrap repetitions in the above simulations, we also conducted a sensitivity analysis to evaluate how robust the results were when varying the number of bootstrap repetitions, adding $B = 500$ and $B = 2000$. The results are summarized in Table S2. A larger B resulted in a slight decrease in standard deviation of the interval length, but the impact was minimal on performance in terms of coverage, and prediction interval length (Andrews and Buchinsky, 2000).

We also conducted a simulation study when the censoring times C depended on covariate $\mathbf{X}$ (more details on data generation in [Supplementary Material](#)). As shown in Table 2, when the censoring distribution was correctly accounted for, the proposed method achieved the targeted 90% coverage probability across all working models. Widths of the intervals are slightly wider than their counter parts when C was independent of $\mathbf{X}$. It is somewhat surprising that even when the censoring times depend on $\mathbf{X}$ but are ignored in the estimation procedure, the coverage rate could still maintain around 90% when T followed the log-normal distribution. There was a moderate undercoverage (around 85%) when T followed the Weibull distribution and the censoring distribution was mis-specified. We have also considered a setting with a high dimension of covariates with $p = 100$

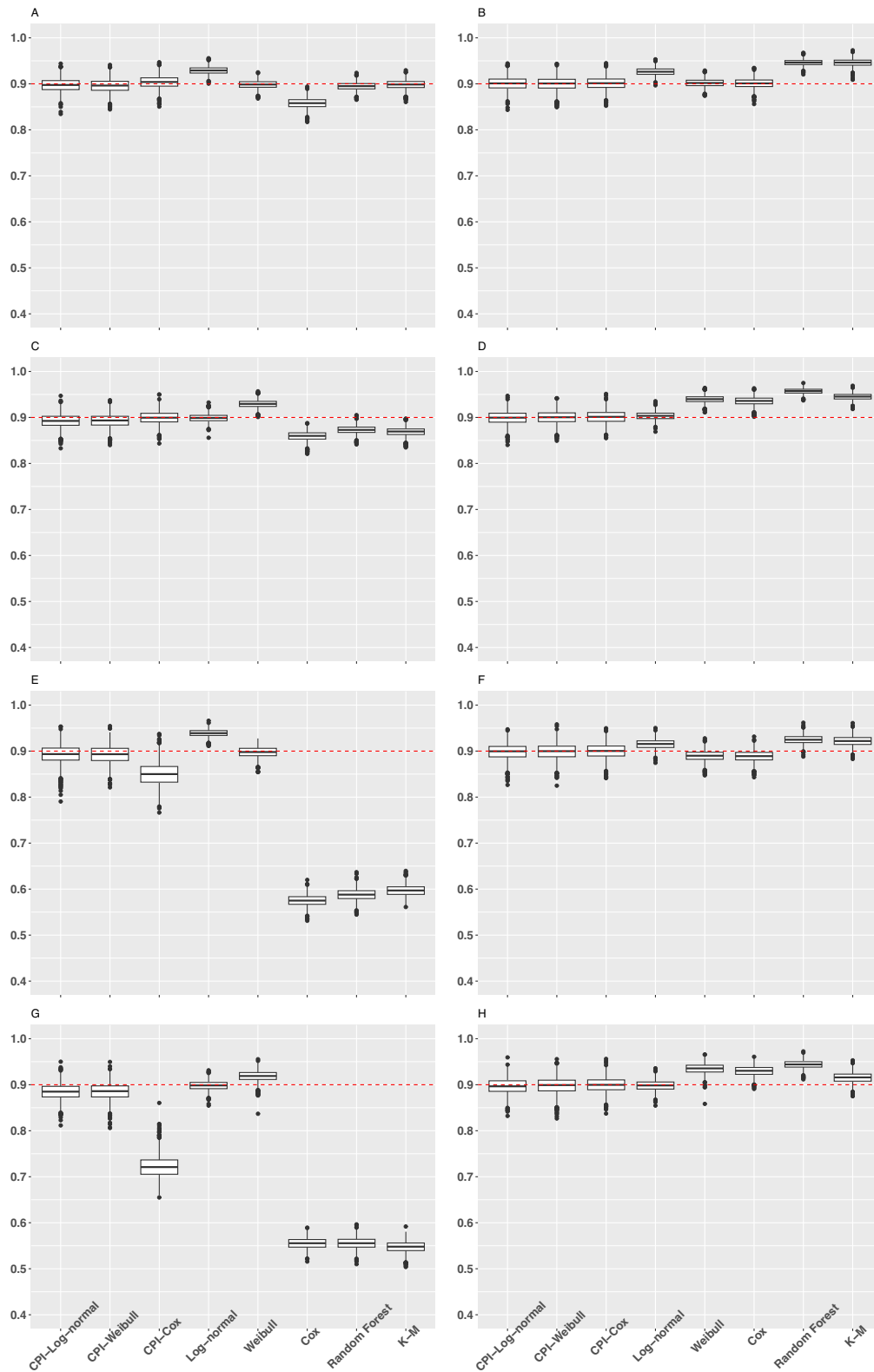


FIGURE 1 Two-sided empirical 90% coverage of the failure times where data were from (A) Weibull model, censoring rate = 15%; (B) Weibull, $\min(\text{failure times}, \eta)$, censoring rate = 15%; (C) log-normal, censoring rate = 15%; (D) log-normal, $\min(\text{failure times}, \eta)$, censoring rate = 15%; (E) Weibull model, censoring rate = 50%; (F) Weibull, $\min(\text{failure times}, \eta)$, censoring rate = 50%; (G) log-normal, censoring rate = 50%; and (H) log-normal, $\min(\text{failure times}, \eta)$, censoring rate = 50%.

TABLE 2 Empirical 90% coverage of the failure times when C depends on $\mathbf{X}$: when this dependence is ignored in the estimation using $G(t)$ and when it is properly considered using $G(t|\mathbf{X})$.

Distribution	Setting	Method	Cov	Length	SD
Weibull	$G(t \mathbf{X})$	CPI-log-normal	0.90	21.24	2.96
		CPI-Weibull	0.90	21.18	2.85
		CPI-Cox	0.90	21.67	2.77
Weibull	$G(t)$	CPI-log-normal	0.84	16.72	1.32
		CPI-Weibull	0.84	16.74	1.28
		CPI-Cox	0.85	17.20	1.64
Log-normal	$G(t \mathbf{X})$	CPI-log-normal	0.90	70.64	19.59
		CPI-Weibull	0.90	70.79	20.48
		CPI-Cox	0.90	43.83	25.08
Log-normal	$G(t)$	CPI-log-normal	0.90	68.07	15.39
		CPI-Weibull	0.90	68.34	16.95
		CPI-Cox	0.91	41.41	16.20

(details are in Table S3 of [Supplementary Material](#)). The proposed approaches effectively maintained the specified coverage probability.

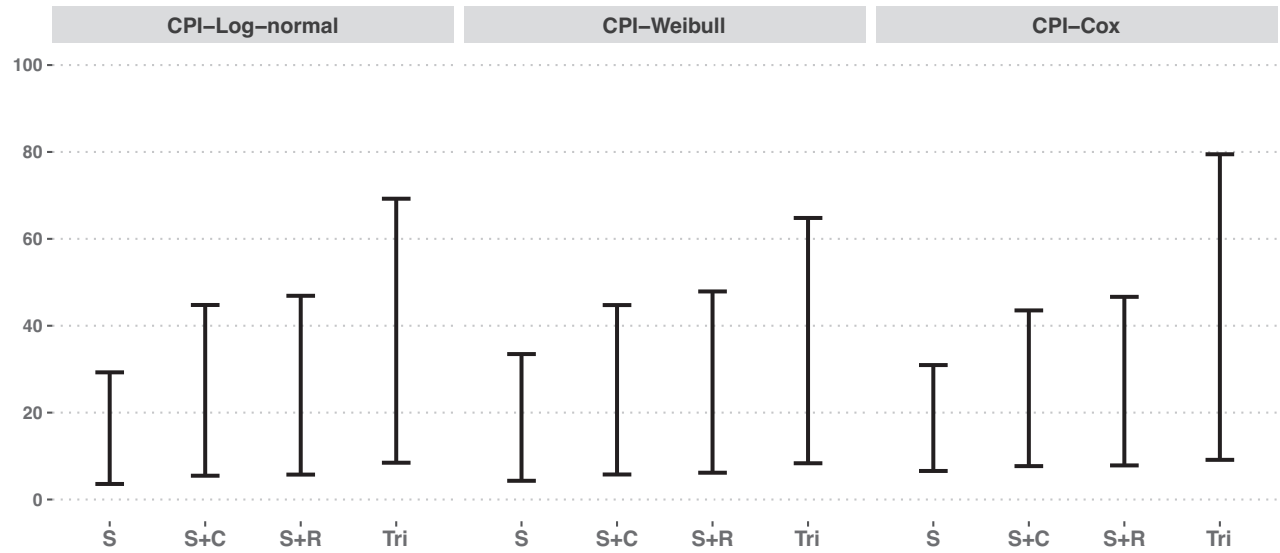
6 APPLICATION TO THE BREAST CANCER DATA FROM THE NCDB DATABASE

Using the proposed methods, we estimate the distribution-free CPIs for non-metastatic IBC patients' survival outcome given their tumor characteristics and initial treatment received. A total of 10 197 female patients with clinical diagnosis of stage IIIB or IIIC IBC breast cancer who underwent surgical treatment during 1998–2010 were identified from the NCDB database. Among these patients, 4217 had complete information on death or censoring times, as well as covariates, including age, race, Charlson–Deyo Score (CDS), tumor grade, nodal stage, and treatment. Current IBC treatment guidelines recommend neoadjuvant chemotherapy followed by modified radical mastectomy and post-surgery radiation therapy, referred as trimodality treatment (Dawood et al., 2011). The above baseline risk factors are shown in Table S5 in the [Supplementary Materials](#). We randomly split patients into a training set (70%) and a testing set (30%) and calculated the coverage of the proposed method. As the actual failure time was only observed for the uncensored subjects, the algorithm for splitting conformal prediction to survival outcomes was employed to validate the overall coverage probability of CPIs before providing the estimated CPIs for patients with different risk factors. More detailed descriptions of the validation algorithm, including the splitting method and simulation results, were provided in [Supplementary Material](#). We repeated the process of random splitting 100 times, and the results are summarized in Table S6. The validation algorithm confirmed that the proposed methods (CPI-log-normal, CPI-log-Weibull, and CPI-Cox) achieved the specified coverage probability of 90% (5th–95th percentile, 8.5–83.7 months), (8.3–82.8 months), and (8.3–79.5 months), respectively, for the 3 CPIs. The estimated CPIs were comparable under the 3 working models overall.

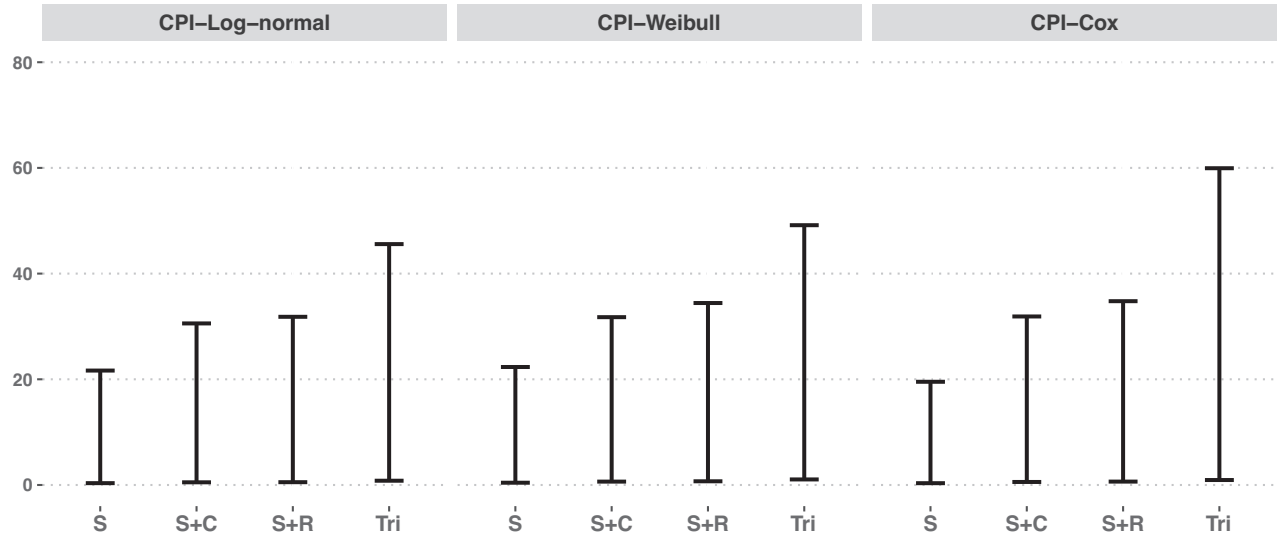
We then estimated the CPIs for IBC patients and examined the impact of race, CDS, tumor grade, nodal stage, and especially treatment by varying the levels of variables. The conformal pre-

diction intervals for patients with various profiles were estimated and are shown in Figure 2(a) and Table S7. For example, it is of interest to estimate the 90% prediction interval for the survival time of a 55-year-old White woman with IBC, no comorbidities, nodal stage N2, and a poorly differentiated tumor. Figure 2(a) and Table S7 show that if she receives the trimodality treatment, her 90% predictive survival time is in (8.5, 69.2), (8.3, 64.8), and (9.1, 79.5) months under the log-normal, Weibull, or Cox working model, respectively. For the same patient, if she decides to receive surgery alone, her 90% predictive survival time is in the range of (3.6, 29.3), (4.3, 33.5), and (6.6, 31) months under the log-normal, Weibull, or Cox working model. Under the same treatment (eg, trimodality), patients with lower CDS scores, well-differentiated or moderately differentiated tumors (compared to poorly- or undifferentiated tumors), and a lower nodal stage showed a higher low bound and a much higher upper bound, as shown in Table S7. We recommend to use CPIs under the parametric working models given the heavy censoring rate (65%) in this dataset. As shown in Figure (S2), it is important to note that the upper bounds of the 90% prediction intervals differ significantly across treatment options, with differences of around 3–4 years. In contrast, the lower bounds vary by only 1–4 months, suggesting that the 1-sided conformal interval provides limited information on the lower end of survival predictions. These results highlight the substantial potential for improved survival with trimodal treatment compared to surgery alone, emphasizing the importance of considering the upper bound in survival predictions. Reliable patient-specific survival time predictions are crucial in making informed, personalized treatment decisions based on individual tumor characteristics.

Given that a patient has survived for 24 months after her initial IBC diagnosis and treatment, it is of interest to predict her remaining survival time based on her tumor characteristics and treatment received as shown in Figure 2(b). For example, a 55-year-old White patient with CDS (0), PD, and N2 who received trimodality treatment had a 90% predictive interval of (0.8, 45.6) and (1.1, 49.2) months of remaining survival using the CPI-log-normal and CPI-Weibull approaches, respectively. This was compared to (0.3, 21.7) and (0.4, 22.3) months for the same subject if she had surgery alone, (0.5, 30.6) and (0.6, 31.8) months if she had surgery and chemotherapy, and (0.5, 31.8) and (0.7, 34.4) months if she had surgery and radiation.



(A) Predicted 90% coverage of the failure times in month for a 55-year-old white patient without any co-morbidity condition, nodal stage N2, and poorly-differentiated tumour.



(B) Predicted 90% coverage of the remaining failure times in month for a 55-year-old white patient without any co-morbidity condition, nodal stage N2, and poorly-differentiated tumour who has survived 24 months after the initial IBC diagnosis.

FIGURE 2 Predicted coverage of the failure times (A) or the remaining failure times (B) for a patient based on a given profile. The treatment options are: surgery alone (S), surgery plus chemotherapy (S+C), surgery plus radiation (S+R), and surgery, chemotherapy, and radiation (Tri).

7 CONCLUDING REMARKS

While conformal inference has been successfully used in uncensored data and machine learning covariate shift problems, this method faces new challenges in predicting right-censored survival outcomes. One key difference between our proposed method and the approach by Candès et al. (2023) lies in the censoring mechanism. While their method assumes type I censoring, we propose a bootstrap method to construct the CPI for survival data subject to general right censoring.

We consider the joint distribution of $(T, \mathbf{X})$ in the pivot, $S(T | \mathbf{X}; \theta)$, instead of the conditional distribution in traditional in-

ference. A step of resampling paired data $(T_i, \mathbf{X}_i)$, where T_i is uncensored, from this consistently estimated distribution, represents a unique approach to constructing the conformal scores asymptotically. By using bootstrap replicates, the lower and upper quantiles for the conformal scores can be calibrated from the original data, taking into account the variation of both T and $\mathbf{X}$. Given that the pivot $S(t | \mathbf{X}; \theta)$ is a monotone function of t , one can easily calculate the predictive interval for the future survival time T_{n+1} given a set of covariates. Kernel smoothing can be a robust method for estimating the conditional distribution of T given $\mathbf{X}$, but it suffers from the curse of dimensionality,

especially when $\mathbf{X}$ is high-dimension or consists of continuous variables. This makes the conformal prediction approach preferable. Moreover, we conjecture that our methods may extend to cases where the dimension of $\mathbf{X}$, p , increases with n , as supported by literature on conformal inference for uncensored data (Chernozhukov et al., 2014; Lei et al., 2018). This warrants further investigation in a future study.

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SUPPLEMENTARY MATERIALS

Supplementary material is available at *Biometrics* online.

Web Appendices and Tables referenced in Sections 3, 4, 5, and 6, along with the code, are available with this paper at the Biometrics website on Oxford Academic. The programming code and simulated dataset are available in the Github: <https://github.com/JPiao7u089/CPI-bootstrap.git>.

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CONFLICT OF INTEREST

None declared.

DATA AVAILABILITY

The National Cancer Database used in the paper cannot be shared publicly due to our agreement with the National Cancer Database for patients' privacy protection. If readers are interested in obtaining access to the National Cancer Database for a specific study, please contact National Cancer Database directly at ncdb_puf@facs.org, to facilitate a data agreement with National Cancer Database.

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